DATA ANALYSIS PROCESS · SESSION MODULE

Exploratory Data Analysis

Understanding your data through Univariate, Bivariate & Multivariate analysis

What & Why of EDA | Univariate Analysis | Bivariate & Multivariate Analysis | Assumptions & Checks

Prepared for: Data Analysis Process Curriculum

01

What is EDA?

Getting to know your dataset before you model it

Definition

Exploratory Data Analysis (EDA) is the process of investigating and summarizing a dataset's main characteristics using statistical summaries and visual methods, before applying formal modeling or hypothesis testing.

*Coined by statistician **John Tukey** in 1977 as a philosophy for letting data reveal its own structure, patterns, and anomalies.*

Visual Methods

Charts & plots reveal shape, spread, relationships

Statistical Summaries

Mean, median, std, quartiles, skewness

Data Quality Checks

Missing values, duplicates, inconsistent entries

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Why Do We Use EDA?

The value it adds before modeling begins

Detect Outliers & Anomalies

Spot unusual values that could distort statistics or models

Understand Relationships

Reveal how variables interact and influence each other

Validate Assumptions

Check normality, linearity & other model prerequisites early

Assess Data Quality

Identify missing, duplicate, or inconsistent records

Guide Feature Engineering

Inform which features & transformations to use

Build Data Intuition

Develop informed, evidence-based decisions before modeling

03 Objectives of EDA

What a thorough exploration sets out to achieve

1. Summarize the dataset's main characteristics (shape, size, types)
2. Identify the underlying distribution of each variable
3. Detect outliers, anomalies & data entry errors
4. Uncover relationships & patterns between variables
5. Check assumptions required by statistical models
6. Decide on cleaning, transformation & feature engineering steps

PART ONE

Univariate Analysis

Examining one variable at a time — its distribution, spread & shape

04 Univariate Analysis — Overview & Steps

Analyzing the distribution of a single variable

Steps for Univariate Analysis

1. Identify the variable type — categorical or numerical
2. Compute summary statistics (count, mean, median, mode, std, quartiles)
3. Choose the plot suited to the variable type
4. Visualize the frequency / distribution shape
5. Detect skewness, outliers & unusual spikes
6. Record observations to guide cleaning & feature decisions

Categorical Column

Bar Plot · Pie Plot

Numerical Column

Histogram · KDE · Box Plot

05 Univariate — Categorical Column

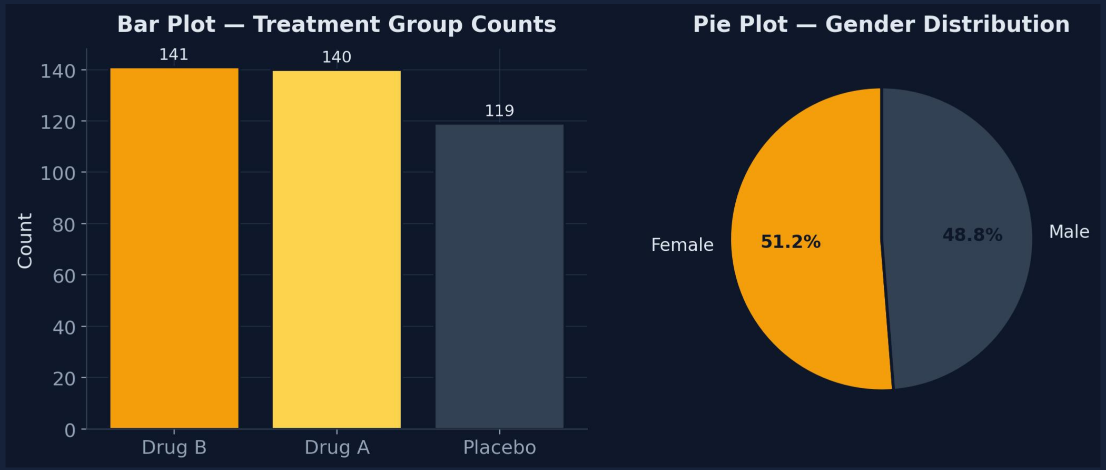
Frequency & proportion of category labels

Steps: count unique categories with value_counts() → plot frequency (Bar) → plot prop

```
# Univariate — Categorical column
import matplotlib.pyplot as plt

counts = df['Treatment'].value_counts()

# Bar plot
counts.plot(kind='bar', color='#F59E0B')
plt.title('Treatment Group Counts')
plt.show()

# Pie plot
```

Bar Plot — best for comparing frequency/count across categories.

Pie Plot — best for showing proportion of a whole (use with few categories).



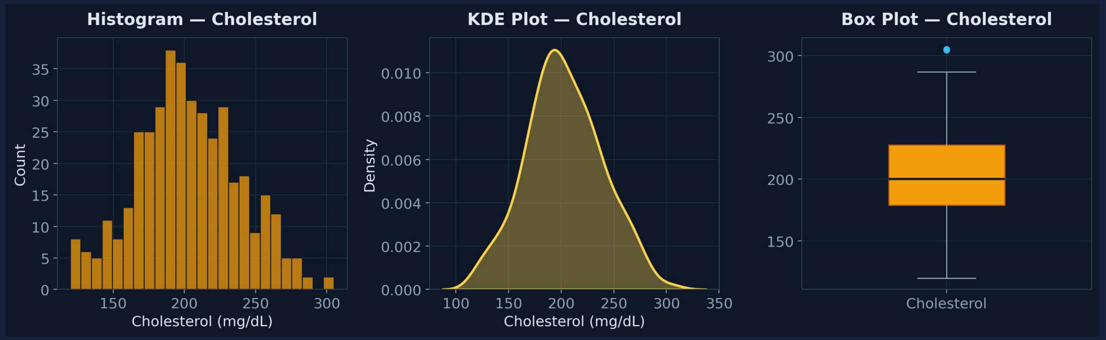
06 Univariate — Numerical Column

Distribution shape, central tendency & spread

Steps: run describe() for summary stats → plot Histogram for frequency bins → plot KDI

```
# Univariate — Numerical column
import seaborn as sns

df['Cholesterol'].describe()

# Histogram
sns.histplot(df['Cholesterol'], bins=25)

# KDE plot
sns.kdeplot(df['Cholesterol'], fill=True)

# Box plot
sns.boxplot(y=df['Cholesterol'])
```

Hist shape/skew · **KDE** smooth density · **Box** median, IQR & outliers



PART TWO

Bivariate & Multivariate Analysis

Examining relationships between two or more variables

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Bivariate & Multivariate — Overview & Steps

Two variables vs. three or more, analyzed together

Bivariate Analysis
Studies the relationship between exactly two variables (e.g. Age vs Cholesterol).

Multivariate Analysis
Studies relationships among three or more variables simultaneously (e.g. Age, BMI & Cholesterol together).

General Steps

1. Identify the type of each variable pair (num-num, cat-num, cat-cat)
2. Select the plot & statistic suited to that combination
3. Compute correlation / association measures
4. Visualize the relationship & interpret strength/direction
5. Check assumptions (linearity, homoscedasticity, independence)

Variable-Pair → Plot Mapping

Num – Num	Scatter Plot, Pair Plot, Correlation Heatmap
Cat – Num	Box Plot, Violin Plot, Bar Plot of Means
Cat – Cat	Crosstab Heatmap, Stacked/Grouped Bar, Countplot

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Numerical — Numerical

Scatter relationships & correlation strength

Steps: plot a Scatter plot to visualize the relationship → compute Pearson correlation →

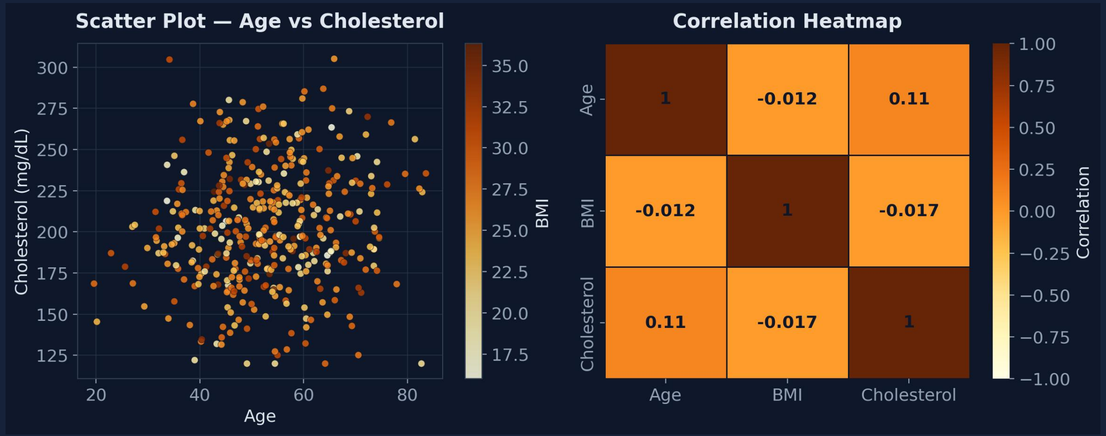
```
# Numerical vs Numerical
import seaborn as sns

# Scatter plot
plt.scatter(df['Age'], df['Cholesterol'])
plt.xlabel('Age'); plt.ylabel('Cholesterol')

# Correlation matrix
corr = df[['Age','BMI','Cholesterol']].corr()

# Heatmap
sns.heatmap(corr, annot=True, cmap='YlOrBr')
plt.show()
```

Pair Plot (sns.pairplot) extends this to every numeric pair at once — useful for a full multivariate first look.



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Categorical — Numerical

Comparing a numeric variable across groups

Steps: group data with groupby() → compare summary stats per group → plot Box/Violi

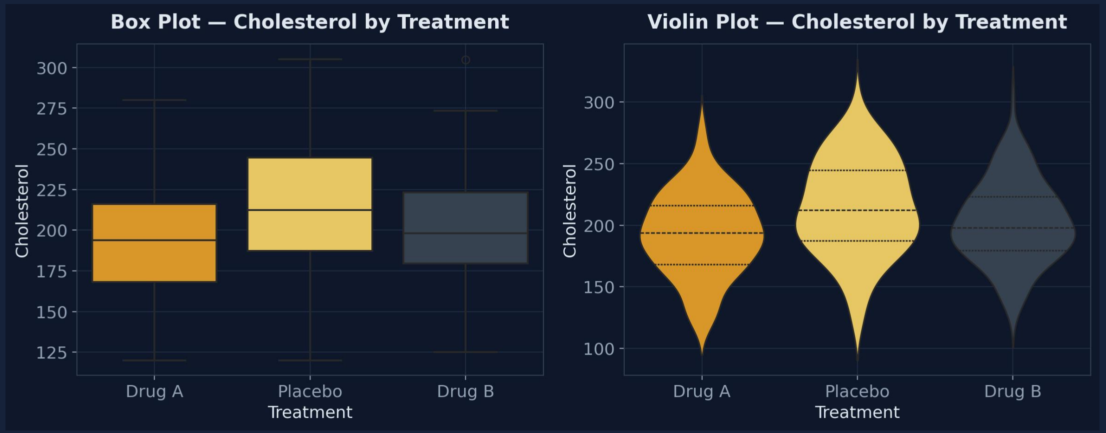
```
# Categorical vs Numerical
import seaborn as sns

df.groupby('Treatment')['Cholesterol'].mean()

# Box plot per group
sns.boxplot(x='Treatment', y='Cholesterol',
            data=df)

# Violin plot per group
sns.violinplot(x='Treatment', y='Cholesterol',
               data=df)
plt.show()
```

Box compares medians/IQR · **Violin** adds full distribution shape per category.



10 Categorical — Categorical

Association between two category variables

Steps: build a Crosstab / contingency table → visualize with a Heatmap → plot a Stacked

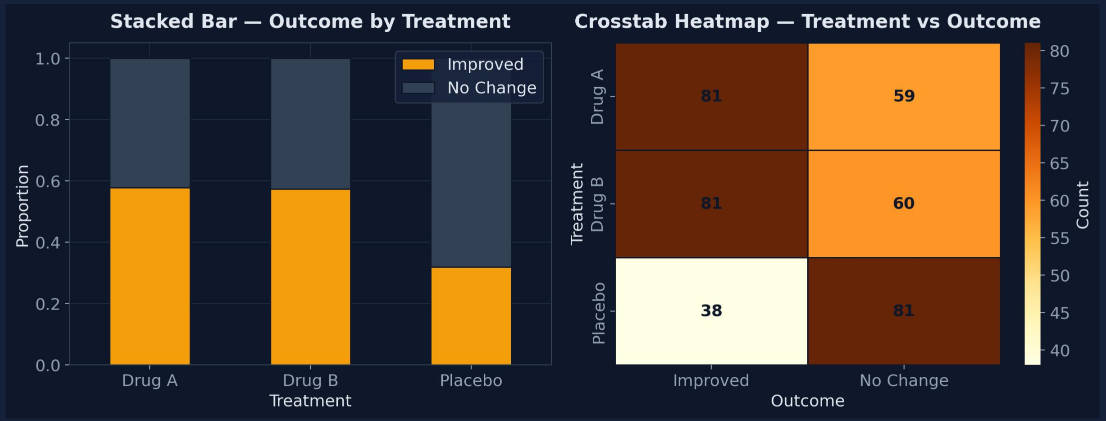
```
# Categorical vs Categorical
import pandas as pd, seaborn as sns

ct = pd.crosstab(df['Treatment'],
                 df['Outcome'])

# Heatmap of counts
sns.heatmap(ct, annot=True, fmt='d',
            cmap='YlOrBr')

# Stacked bar of proportions
ct.div(ct.sum(1), axis=0).plot(
    kind='bar', stacked=True)
```

Chi-Square Test (scipy.stats.chi2_contingency) formally checks whether two categorical variables are statistically associated.



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Assumptions & Checks During EDA

What to validate before trusting the analysis

Normality

Histogram/QQ-plot or Shapiro-Wilk test on numeric variables

Linearity

Scatter plots to confirm linear (not curved) relationships

Homoscedasticity

Residual/spread should stay constant across groups

Independence

Observations shouldn't influence one another

Multicollinearity

Check correlation matrix / VIF among numeric predictors

Missing & Outlier Impact

Assess how nulls/outliers affect distributions & stats

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Key Takeaways

EDA in one page

Univariate

- Analyze one variable at a time
- Categorical → Bar, Pie
- Numerical → Hist, KDE, Box

Bivariate / Multivariate

- Analyze relationships between 2+ variables
- Num-Num → Scatter, Heatmap
- Cat-Num → Box, Violin
- Cat-Cat → Crosstab, Stacked Bar

Assumptions

- Normality & linearity
- Homoscedasticity & independence
- Watch multicollinearity & outliers

Good EDA turns raw data into confident, evidence-based modeling decisions.